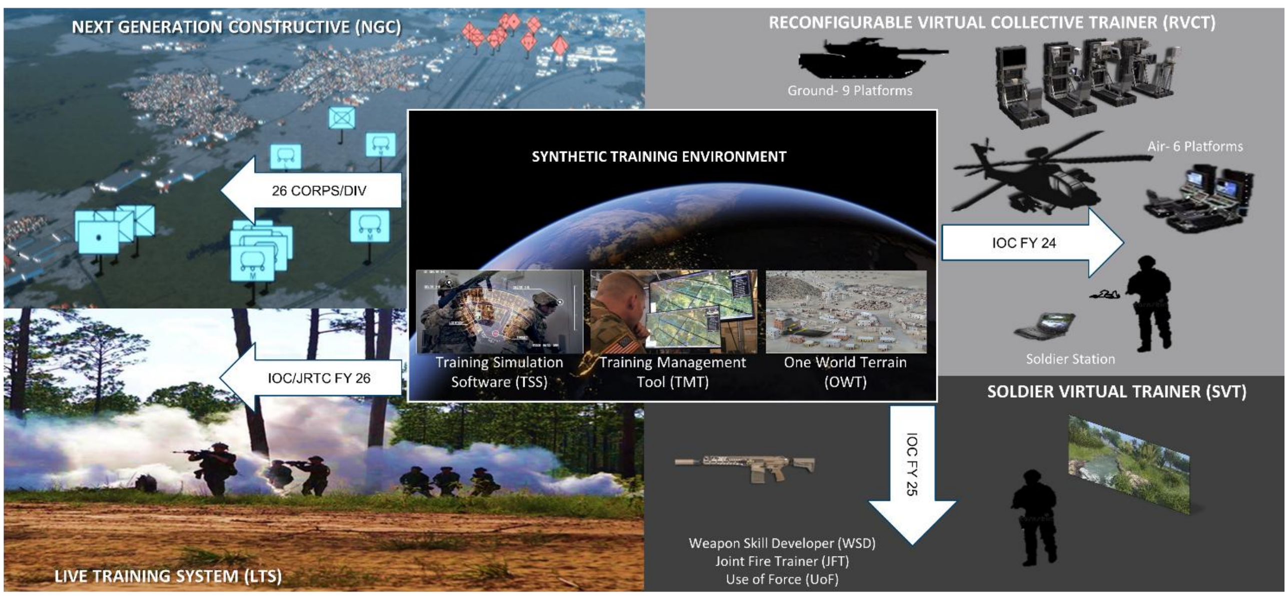
IO-002: Next Generation Constructive (NGC) Acceleration to include Synthetic Training Environment Software (STE-SW)

1. **STE Overview**

The Army's future training capability is the Synthetic Training Environment (STE). The STE will be a single, converged training system that provides a training environment, in which units from Soldier/Squad through Army Service Component Command (ASCC) conduct both collective and constructive training, see Figure 1-1 below.



The STE is envisioned to converge Live, Virtual, and Constructive training environments into one holistic system. STE will utilize one software, one terrain, and one training tool across multiple interfaces. This training capability will enable Army units and leaders to conduct realistic multi-echelon / multi-domain combined arms maneuver and mission command training, increasing proficiency through repetition.

STE provides a synthetic environment to represent and enable rapid generation of the operational environment (OE). Scalable immersive collective trainers provide the commander with multiple training capability options to train operational tasks at the point of need (PoN) with around-the-clock accessibility. The STE operates in connected and disconnected modes (for training under limited or degraded network conditions) and is embedded within the Common Operating Environment (COE). The STE leverages the Army Network (enterprise and tactical) to either host or deliver capabilities and provides intuitive, composable applications and services that enable embedded training with mission command workstations and select platforms. Future Live training systems will interface with the STE to incorporate instrumentation, ranges, targets, and ammunition into the synthetic environment. NGC will expand training of large-scale Brigade Combat Teams (BCT) through ASCC. STE-SW is comprised of the Training Simulation Software (TSS), Training Management Tool (TMT), One World Terrain (OWT), and NGC.

1.1 Background & Problem Statement

The Army currently employs a collection of outdated, cyber-vulnerable, and inefficient live, virtual, constructive, gaming (LVCG) collective training systems that form the Integrated Training Environment (ITE) which is not available at the PoN. These systems do not replicate the complex challenges of the current and future OE nor are they Agile enough to allow Warfighters to train multiple iterations prior to mastering training in the live training environment. As we set the foundation for a MDO capable force in 2028, these critical capability gaps degrade Warfighter training readiness and performance placing our Nation's defense at risk.

The Army has prioritized closing its constructive capability gap in training large scale combined arms collective training to meet the readiness of the Army Operating Concept (AOC), MDO, as well as integrated and collaborative training requirements with Joint Forces and Multi-National Partners. Many of these current gaps require significant artificial manual injects/white cards across the physical, information, and human dimensions that only increases the significant overhead, complexity, and integration for combined arms staff training with Joint & Mission partners. Coupled with this, current constructive training capabilities lack a cohesive and modular architecture and do not rapidly and continuously integrate/deliver software intensive capabilities employing cloud native technologies. This leads to increased waterfall development timelines and legacy manual block upgrades that are not aligned with Army software modernization directives.

2. Desired Objectives

2.1 STE-SW will provide modern, realistic collective and constructive training for Soldier/Squad through ASCC with increased capabilities for Commanders / Soldiers to train and close the ITE limitations without creating new gaps (i.e., Close Combat Tactical Trainer [CCTT]; Aviation Combined Arms Tactical Trainer [AVCATT]; and Games for Training [GFT] and a constructive staff and Commander training capability). The STE system enables the Unit/Staff to train offense, defense, and stability support operations realized through the CATS to set the foundation for an MDO capable force.

2.2 NGC will enable the Army to close command and staff training gaps across the competition continuum within a MDO environment from soldier to ASCC/theater level. The delivery of NGC requires it to leverage the Synthetic Training Environment – Software (STE-SW) foundation, to ensure a single simulation environment scaled across the live, virtual, and constructive domains. STE-SW must be that is further augmented, extended, and integrated to achieve Army training readiness needs to enable training in accordance with the full breadth of FM3.0. The STE ecosystem enables the Army to fully realize standard of training proficiency.

2.3 STE-SW should be capable of being developed, distributed, and maintained within a government managed cloud infrastructure. NGC should employ best of breed approaches that enable the M&S community to test, experiment, build models, and validate capability. The STE must have the ability to evolve and adapt with technology allowing for the replication/representation of current and future force structure, weapons effects, warfighting functions, Joint, Interagency, and Multinational (JIM) capabilities, human interaction, dense urban environment, and near-peer, hybrid, and asymmetric friendly and threat capabilities. The STE must be on the forefront of technology and should be able to rapidly incorporate emerging technologies to streamline development, automate features and functions, reduce overhead, and increase realism within the training environment. The STE will be capable of training units across the full range of Military Operations in multiple domains (maritime, land, air, space, and cyberspace) and dimensions (physical, information, and human). The STE will be capable of training units across the full range of Competition and Conflict Continuum.

2.4 STE-SW will provide and continuously integrate/develop common software services/capabilities across the STE ecosystem consisting of Live Training Systems (LTS), Reconfigurable Virtual Collective Trainer (RVCT), and Soldier Virtual Trainer (SVT).

2.4.1 TSS is the simulation engine that generates an immersive operational environment to enable virtual collective training and constructive staff training.

2.4.2 TMT is a common and intuitive tool that allows Soldiers at every level to plan, prepare, execute, and assess training objectives and scenarios.

2.4.3 OWT provides 3D high fidelity terrain to support realistic and immersive training and rehearsals.

2.5 Using the modern software acquisition pathway, STE-SW leverages agile software development practices with iterative roadmap capability deliveries providing capability aligning to Army software modernization directives. STE SW rapidly integrates Government-Off-the-Shelf (GOTS), Commercial-Off-The-Shelf (COTS), and open-source technologies, solutions, standards, and products that align to a Modular Open Systems Approach (MOSA) architecture. STE-SW must be forward leaning, taking advantage of modern APIs, modern protocols, use of artificial intelligence, automation of development, and automation of constructive simulation tasks (i.e. elimination of white cards during exercises). Defined within the STE-SW Platform Development Kit (PDK) that has been made available to industry through several iterations, STE enables extensibility, pluggability, and composability. Note that the STE PDK contains Controlled Unclassified Information (CUI), responding vendors must adhere to USG cybersecurity policies, and must be US-Owned companies with an active Commercial and Government Entity (CAGE) code for security verification purposes.

2.6 The U.S. Army is looking for solutions to train soldier through theater level commanders, staffs, and subordinate units in all warfighting functions (Movement & Maneuver, Fires, Protection, Sustainment, Intelligence) that fall within multidomain operations (Land, Air, Maritime, Space, Cyber & Information). Additionally, the US Army is looking for solutions in training enablers, e.g. infrastructure.

2.6.1 "Movement and Maneuver, Fires, Protection" contains, but is not limited to, simulation of: Direct fire, indirect fires coordination; cannon, mortar, and rocket; air defense artillery; aviation & air base operations; ground base operations; offensive and defensive maneuvers; reactive behaviors; engineering (digging, breach, bridging), displaced persons management.

2.6.2 "Sustainment" contains, but is not limited to, simulation of supply point operations; transportation of material, ammunition, vehicles, aircraft; maintenance repair time and replacements; medical operations; personnel reception, staging, onward-movement & integration, Logistics Status (LOGSTAT) and Personnel Status (PERSTAT); report generation (e.g. SPOT, Battle Damage Assessment (BDA), Situation Report (SITREP)).

2.6.3 "Intelligence" contains, but is not limited to, simulation of: NGC Intel Core Model design; Named Area of Interest (NAI) collection reporting; Virtual Terrestrial Electronic Warfare (EW); Moving Target Indicator (MTI) generation; Positioning, Navigation, and Timing (PNT) satellites and jamming; Radio Frequency (RF) propagation models; Geospatial Intelligence (GEOINT) capabilities that include Imagery Intelligence (IMINT) products: Full Motion Video (FMV), Synthetic Aperture Radar (SAR), and Ground MTI (GMTI) modeling from aircraft and satellites; Signals Intelligence (SIGINT) capabilities that include: Aerial-based and Space-based Electronic Intelligence (ELINT) reporting and Aerial-based Communication Intelligence (COMINT) reporting; Human Intelligence (HUMINT) capabilities that include: Army sources consisting of Enemy Prisoners of War (EPW), Military Source Operations (MSO), and Internally Displaced

Persons (IDP), alongside Joint, Interagency, Intergovernmental, and Multinational (JIIM) sources; satellite models; communication jamming; constructive electromagnetic attack (EA).

2.6.4 "Multi-Domain Operations" contains, but is not limited to, simulation of: space research development; cyber effects research development; space & global positioning systems representation; cyber effects research development; space intelligence; space weather; hypersonics; energy weapons, electronic warfare; nonlethal and information warfare; contested logistics and logistical flexibility; joint targeting; cross domain fires, robotics, and artificial intelligence.

2.6.5 "Infrastructure" Approaches and paths to simplify delivery of scaled NGC capabilities over more traditional legacy constructive simulation approaches that contain but is not limited to: enterprise cross-domain solutions (CDS) analysis; CDS architecture refinement; CDS implementation; CDS risk reduction/testing; cloud cost/benefit analysis; and cloud onboarding. Infrastructure also includes effective Mission Command (MC) feeds from simulation with appropriate feeds, overlays, and coalition interoperability. Infrastructure should also support emerging requirements such as Next Generation Command and Control (NGC2) integration and Chemical, Biological, Radiological, Nuclear, and Enhanced conventional weapons (CBRNE) modeling.

3. Requirements

3.1 The requirements defined herein form the basis for all required vendor solutions regarding the design, development, integration, testing, delivery, and product support of STE-SW. Given the possibility of multiple awards under this agreement to multiple vendors, the U.S. Army is looking for solutions in the following capabilities within a teaming, cooperative environment. Vendor capabilities listed below are for illustrative purposes, do not represent an exhaustive listing, and are subject to change within the sprint development process. Solution papers may choose to address one, some, or all of the following capabilities. All proposed solutions should consider program management plans, communication, integration efforts, configuration management, and all other necessary efforts needed to meet each requirement. Each vendor must provide a product roadmap that identifies anticipated features across each six-month minimum viable product (MVP). Additionally, each vendor must provide an Integrated Master Schedule (IMS) that identifies project management tasks and dependencies, highlighting critical path(s).

3.2 STE-SW Core Services

Effective solutions to address STE software core services will address the vendor's approach to perform as the lead STE SW integrator, which must work with multiple vendors, integrating with all STE software developers to roadmap the overall STE capability. Responsibilities include integrating all STE-SW vendor capability within STE, establishing joint interoperability with service partners and coalition interoperability with allied partners. This includes incorporating "Parking Lot" sessions for key discussions to reach Government consensus and direction. The Lead STE Integrator is also responsible for the integration, testing, leading the cloud migration from IL5 to IL6, and leading the Enterprise Cloud CDS integration efforts. This involves coordinating with ECMA, PEO C3N, and associated government agencies to decompose STE-SW CDS requirements and integrate them through the agile process. This involves aligning Enterprise CDS initiatives with the STE-SW Integrator and capability vendors to ensure proper CDS integration across all STE-SW capabilities. Additionally, the role includes co-developing with all STE-SW vendors to ensure seamless STE integration, planning STPs and unit engagements, providing Help Desk and Operations support for all STE-SW. The STE core services provider should have a plan for enabling government and non-government users to build, test, and integrate capability into the master baseline. This effort reduces risk and cost to the program by enabling stakeholders to onboard, fork, develop, test, and integrate capability much more rapidly and the conditions on the battlefield change.

3.3 STE-SW Test Lead

The vendor shall be responsible for the full range of testing activities associated with planning, coordination, and documentation of the STE system capability and interfaces to include: TSS, TMT, NGC, RVCT, OWT and future STE capabilities. Documentation includes test plan, test procedures, problem tracking reports, etc., all to be stored in a central digital repository. Vendor will be responsible for the full array of testing, from testing new capabilities for verification of completion, verifying problem trouble ticket closure, including system-of-system end-to-end testing. Vendor will be responsible for planning and coordination for all test activities, including major events such as Soldier Touch Points, Operational Demonstrations, etc. The vendor will be encouraged to leverage automation, to the extent possible, to increase efficiencies and the ability for test repetition and decrease defects.

3.4 STE-SW Cyber Lead

The STE-SW Cyber lead shall maintain cyber posture and ATOs for the complete STE-SW system. They shall prepare RMF packages and maintain ATOs for the development environment, on premise training systems, and cloud deployments at unclassified, secret and top-secret levels. The Cyber lead shall support required government compliance processes and tests. They shall maintain the system's cyber posture as new development and modernization by other capability developers occurs. They shall participate in agile development activities of the core software and other capability developers to ensure cybersecurity requirements are considered and proactively addressed. The vendor shall document all IAVA, STIGs, Bulletins, and security control implemented, not implemented, justification for items not implemented, impacts, and mitigations in the POA&M in eMASS. They shall identify, track and allocate system vulnerabilities for remediation by core software and capability developers. They shall maintain the authorizations for required Cross Domain Solutions (CDS).

3.5 Virtual Collective Capability Developer

The Virtual Collective Capability Developer will maintain and enhance the existing STE collective training simulation, enabling the RVCT Ground, Air, and Soldier capabilities within the STE. The Developer will maintain concurrency of the Virtual Collective Capability with the operational platform, including but not limited to platform SW updates and integration with updated platform Mission Command Information Systems (MCIS). The Developer will modernize Virtual Collective Capability, including recommending and implementing approved upgrades to simulation components.

3.7 Movement and Maneuver, Fires Capability Developer

The Movement, Maneuver, Fires Capability Developer is responsible for development of MMF capabilities listed in the “Desired Objectives” section, decomposing and planning MMF requirements into the product backlog, working with other capability vendors and the STE Integrator to ensure MMF requirements traceability aligns with STE requirements managed by the STE Integrator, as well as ensuring new capabilities leverage existing STE architecture. The MMF Capability Developer will deconflict through the STE-SW governance board prioritization and deliver MMF capability through the agile process. They are responsible for the development, integration, testing, and delivery of MMF capability aligned with current Army doctrine and supporting cloud migration from IL5 to IL6. Additionally, the MMF Capability Developer will support STPs and unit engagements.

3.8 Sustainment Capability Developer

The Sustainment Capability Developer is responsible for all aspects of Sustainment Operations (SOps) simulations capability. This includes development of Sustainment capabilities listed in the “Desired

Objectives" section, decomposing and planning SOps requirements into the product backlog, working with other capability vendors, and the STE Integrator to ensure SOps requirements traceability aligns with STE requirements managed by the STE Integrator, as well as ensuring new capabilities leverage existing STE architecture. The Sustainment Capability Developer will deconflict through the STE-SW governance board prioritization and deliver NGC capability to the STE Integrator through the agile process. They are responsible for the development, integration, testing, and delivery of SOps capability into NGC, migrating SOps constructive behaviors aligned with current Army doctrine and supporting cloud migration from IL5 to IL6. Additionally, the Sustainment Capability Developer will support STPs and unit engagements.

3.9 Intel & Electronic Warfare Developer

The Intel Developer is responsible for all aspects of Intel, Electronic Warfare, and Threat simulations capability, leveraging Intelligence models within Intelligence Electronic Warfare Tactical Proficiency Trainer (IEWTPT). This includes development of intel capabilities listed in the "Desired Objectives" section, decomposing and planning intel requirements and NGIC-approved Threat models/requirements into the product backlog. The Lead will work with other capability vendors and the STE Integrator to ensure intel requirements traceability aligns with STE requirements managed by the STE Integrator. They will deconflict through the STE-SW governance board prioritization and deliver intel capability to the STE Integrator through the agile process. Responsibilities also include the integration, testing, and delivery of intel, migrating intel constructive behaviors aligned with current Army doctrine, and supporting cloud migration from IL5 to IL6. Additionally, the Lead will support STPs and unit engagements.

3.10 Cyber Effects Capability Developer

The Cyber Effects Lead is responsible for all aspects of Defensive and Offensive Cyber Operations simulations capability. This includes development of Cyber Operations capabilities listed in the "Desired Objectives" section, decomposing and planning DCO/OCO requirements into the product backlog, leveraging existing capabilities such as cyber capabilities for Live, Virtual, and Constructive systems to enable MDO, to initiate DCO/OCO piloting integration efforts, and working with other capability vendors, and the STE Integrator to ensure DCO/OCO requirements traceability aligns with STE requirements managed by the STE Integrator, as well as ensuring new capabilities leverage existing STE architecture. The CO Capability Developer will deconflict through the STE-SW governance board prioritization and deliver DCO/OCO capability to the STE Integrator through the agile process. Responsibilities also include the development, integration, testing, and delivery of DCO/OCO capability, migrating DCO/OCO constructive behaviors aligned with current Army doctrine, and supporting cloud migration from IL5 to IL6. Additionally, the Lead will support STPs and unit engagements.

3.11 Space Models Capability Developer

The Space Models Capability Developer is responsible for all aspects of Space Models simulations capability. This includes development of Space Model capabilities listed in the "Desired Objectives" section, decomposing and planning messaging format requirements into the product backlog, leveraging existing SMS capabilities where practical, and working with other capability vendors, and the STE Integrator to ensure SMS requirements traceability aligns with STE requirements managed by the STE Integrator, as well as ensuring new capabilities leverage existing STE architecture. The Space Models Capability Developer will deconflict through the STE-SW governance board prioritization and deliver SMS capability to the STE Integrator through the agile process. Responsibilities also include the development, integration, testing, and delivery of space capability and supporting cloud migration from IL5 to IL6. Additionally, the Lead will support STPs and unit engagements.

3.12 Information Operations Developer

Enable the planning of information operations tasks and provide effects and feedback through the simulation. Information Operations should provide intelligence through various open-source platforms, impacting will to fight and protects friendly forces through cyber security, operations security, and counter propaganda. This also includes shaping the environment through political, military, economic, social, information, infrastructure, physical environment, and time (PMESII-PT).

3.13 Geospatial / Terrain and 3D modeling Capability Developer

The terrain developer must deliver an integrated, scalable, and high-fidelity terrain solution to support multi-echelon command and staff training, up to Army Service Component Command (ASCC) within a multi-domain operational environment (MDO). To achieve this, STE-SW requires One World Terrain (OWT) to provide accurate, geospatially precise, and scalable 3D terrain and models that enable realistic operational planning, force maneuvering, and decision-making. STE-SW requires OWT's automated terrain ingestion, rapid terrain updates ensuring synchronized, mission-relevant terrain data. The terrain must support multi-resolution fidelity, allowing synthetic forces and training simulations to dynamically adjust between tactical and strategic viewpoints, while enabling adaptive terrain modifications to reflect evolving operational environments. Additionally, OWT will integrate with mission command systems, simulation engines, and synthetic training environments to enhance the realism of multi-domain targeting, cyber and electronic warfare effects, and operational decision support. OWT must align with STE-SW agile development processes, ensuring that terrain data remains continuously updated, accessible on demand, and adaptable to Army training requirements. This capability is essential for STE-SW's ability to deliver scalable, persistent, and globally representative synthetic training environments, allowing commanders and staffs to conduct realistic large-scale exercises, execute operational planning, and refine decision-making processes across all warfighting domains.

Major Milestones & Decision Points

Major Milestones will be based on the agile approach: utilizing sprints to build Minimum Viable Product-Prototypes.

Follow-On Production: Yes

Upon successful completion of these prototype efforts, the Government reserves the right to award follow-on production efforts via either contract or transaction, without the use of competitive procedures. A follow-on production contract is a possibility if the vendor successfully completes the objectives within the project – defined as successful demonstration of methodologies, technologies, and approaches that supports STE-SW objectives.

Anticipated Security Level of the Prototype Project: Unclassified, Secret Releasable to Coalition Partners, Secret US Only, TS/SCI

Anticipated Funding: Subject to Availability of funds

Certainty of Funding: Subject to Availability of funds

Period of Performance: 36 months

Anticipated Data Rights

Unlimited

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